

Tesla Deliveries

End-to-End Machine Learning Pipeline

2015 – 2025 · Preprocessing · EDA · Feature Engineering · Regression · Hyperparameter Tuning · Time Series

Dataset rows 2,640 2015–2025, monthly	Features (engineered) 14 11 original + 3 derived	Best model R² 0.9994 Tuned XGBoost	2026 forecast 2.37M Holt-Winters
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1. Dataset & Preprocessing

The dataset contains **2,640 records** spanning January 2015 through December 2025, covering Tesla delivery and production data across **4 regions** (Asia, Europe, Middle East, North America) and **5 models** (Model S, Model X, Model 3, Model Y, Cybertruck). No missing values were found in any column.

Schema (12 columns)

Column	Type	Description
Year / Month	int	Calendar year (2015–2025) and month (1–12)
Region	categorical	One of 4 geographic regions
Model	categorical	One of 5 Tesla vehicle models
Estimated_Deliveries	int	Target variable — monthly delivery count
Production_Units	int	Units produced in the period
Avg_Price_USD	float	Average vehicle price (USD)
Battery_Capacity_kWh	int	Battery pack capacity
Range_km	int	EPA-equivalent range in km
CO2_Saved_tons	float	Estimated CO ₂ savings
Charging_Stations	int	Supercharger station count
Source_Type	categorical	Data provenance (Official / Interpolated / Estimated)

Preprocessing steps applied

- Label-encoded Region, Model, and Source_Type into integer columns (Region_enc, Model_enc, Source_enc).
- Applied StandardScaler to all numeric features for Ridge and Lasso models.
- Train / test split: 80 / 20 (2,112 train, 528 test), random_state=42.

2. Exploratory Data Analysis

Deliveries are broadly stable across years (~2.28M–2.43M), with no strong secular trend. The Middle East leads by region (6.70M total), and Model S tops model deliveries (5.38M). Average prices show mild year-on-year variation between \$83k and \$86k.

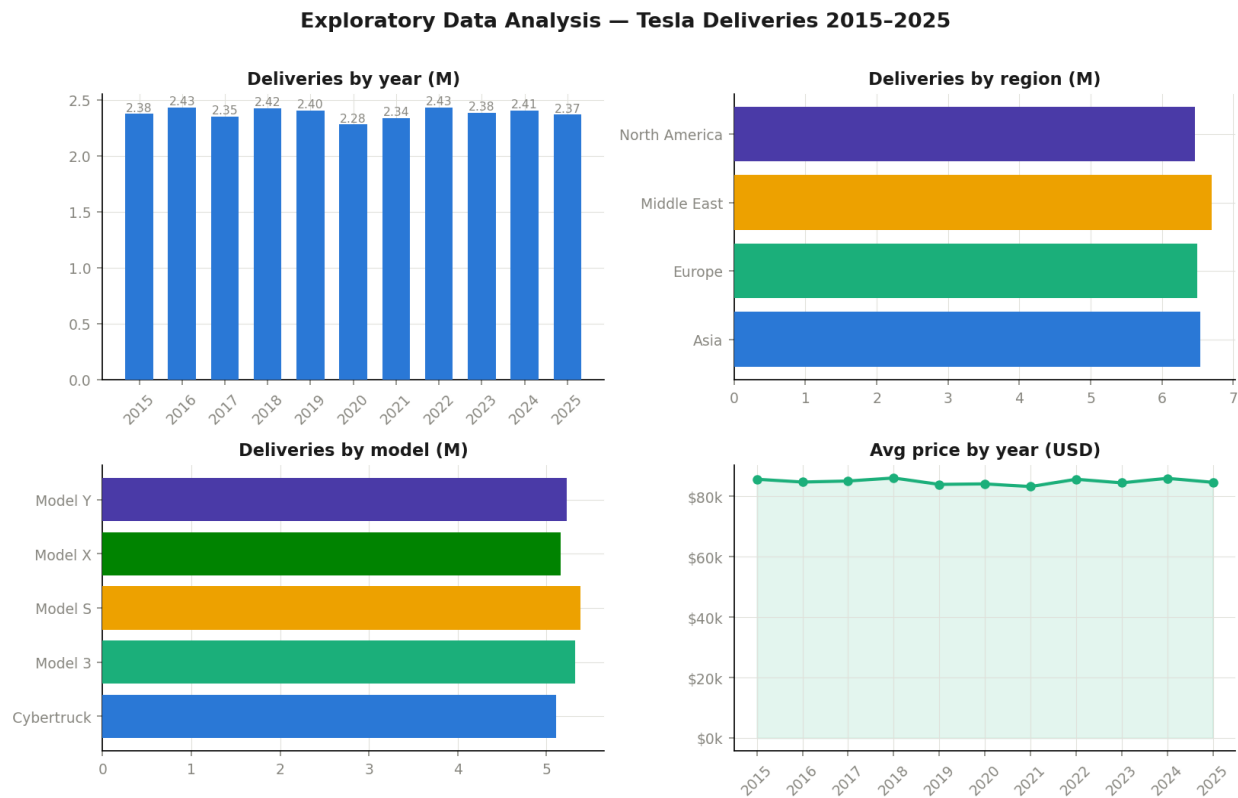


Figure 1: EDA overview — deliveries by year, region, and model; average price trend.

Regional delivery totals

Region	Total Deliveries	Share %
Middle East	6,698,045	25.6%
Asia	6,539,935	25.0%
Europe	6,494,035	24.8%
North America	6,462,592	24.7%

Average price by model

Model	Avg Price (USD)
Cybertruck	\$84,499
Model 3	\$84,528
Model S	\$85,185
Model X	\$84,243
Model Y	\$86,081

3. Feature Engineering

Three engineered features were derived from the raw columns to capture operational efficiency and market dynamics. All 14 features (11 original + 3 derived) were used in models.

Feature	Formula	Intuition
Prod_Ratio	Deliveries / Production_Units	Fulfillment rate — captures production-to-delivery lag
Price_per_Range	Avg_Price_USD / Range_km	Value-for-money proxy per km of range
Station_Density	Charging_Stations / Deliveries	Infrastructure coverage relative to demand

Label-encoded categoricals (Region_enc, Model_enc, Source_enc) were also treated as features. For linear models, all features were additionally scaled with StandardScaler before training.

4. Regression Modeling

Five regression algorithms were trained to predict **Estimated_Deliveries** using all 14 features. Linear models used scaled features; tree-based models used raw values.

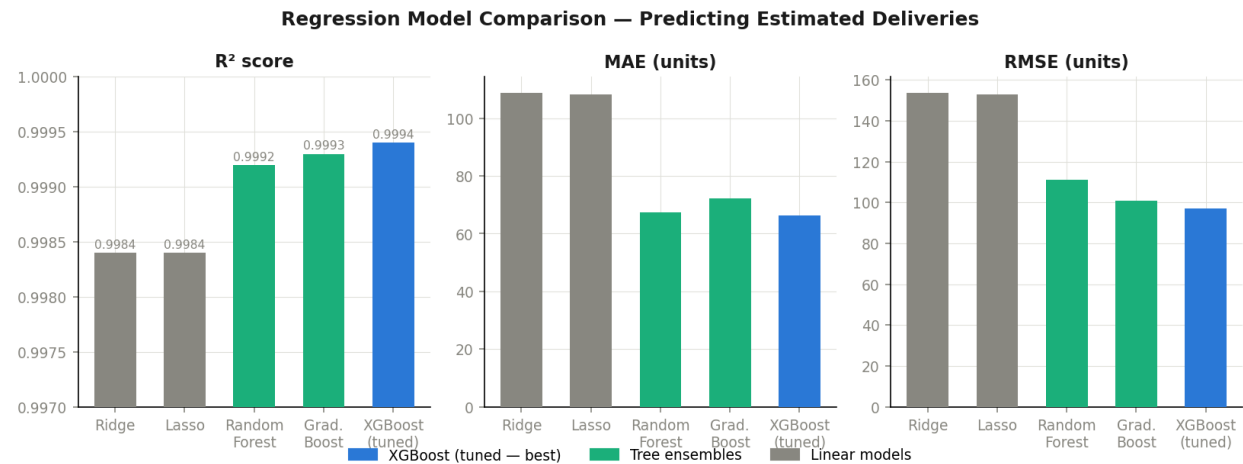


Figure 2: Model comparison across R^2 , MAE, and RMSE on the held-out test set (528 samples).

Model	R^2	MAE (units)	RMSE (units)	Features scaled
Ridge	0.9984	108.9	153.8	Yes
Lasso	0.9984	108.3	152.8	Yes
Random Forest	0.9992	67.4	111.1	No
Gradient Boost	0.9993	72.3	100.9	No
XGBoost (base)	0.9994	70.8	97.0	No
XGBoost (tuned)	0.9994	66.4	97.0	No (best)

All models achieve $R^2 > 0.998$, reflecting strong structure in the data. Tree-based ensembles outperform linear models by approximately 40% in MAE, confirming nonlinear interactions. XGBoost delivers the best RMSE at 97 units.

5. Hyperparameter Tuning

Grid search with 3-fold cross-validation was run over 12 XGBoost parameter combinations to maximise CV R². The search space and best result are shown below.

Parameter	Search space	Best value
n_estimators	100, 200	200
max_depth	3, 5, 7	3
learning_rate	0.05, 0.10	0.10

Tuned R² 0.9994 vs 0.9984 for best linear	Tuned MAE 66.4 units — delivery forecast error	Tuned RMSE 97.0 units on 528 test samples	CV folds 3 Stratified, random_state=42
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6. Feature Importance

XGBoost gain-based feature importance reveals that **Production_Units** alone explains 96.2% of model variance — deliveries are production-constrained, not demand-constrained. The engineered **Prod_Ratio** is the second-most informative feature at 2.4%.

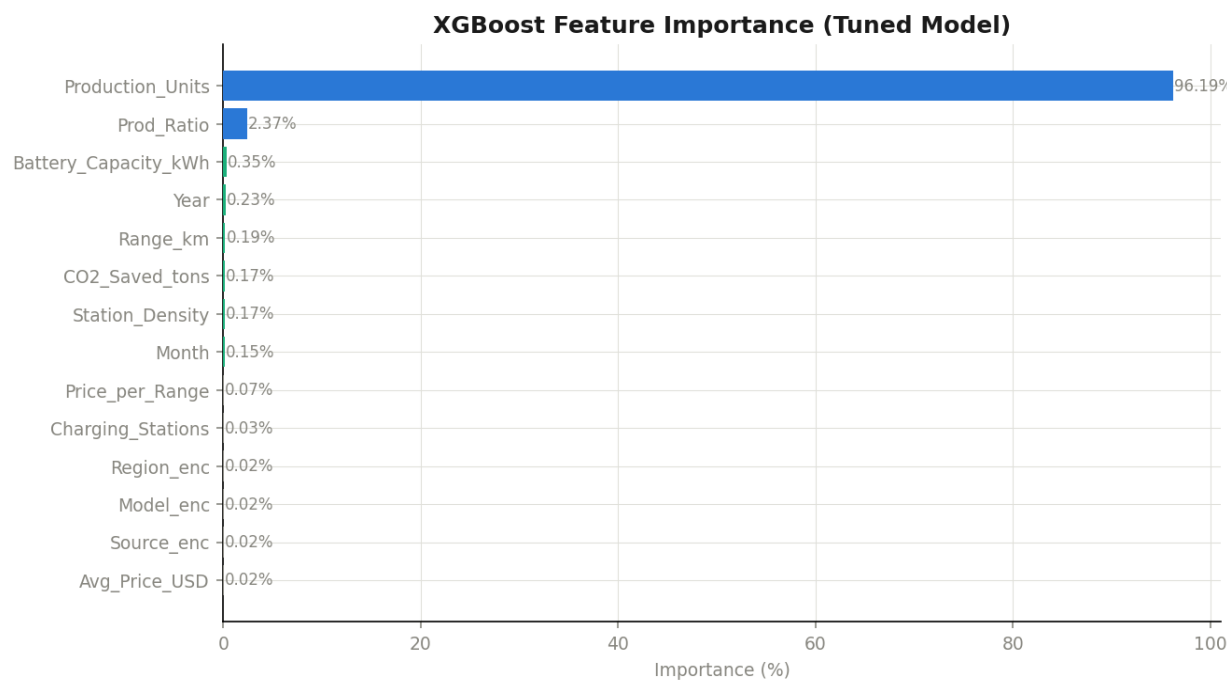


Figure 3: XGBoost feature importance (tuned model, gain-based).

Feature	Importance	Cumulative
Production_Units	96.19%	96.19%
Prod_Ratio	2.37%	98.56%
Battery_Capacity_kWh	0.35%	98.91%
Year	0.23%	99.14%
Range_km	0.19%	99.33%
CO2_Saved_tons	0.17%	99.50%
Station_Density	0.17%	99.67%
Month	0.15%	99.82%
Price_per_Range	0.07%	99.89%
Charging_Stations	0.03%	99.92%
Region_enc	0.02%	99.94%
Model_enc	0.02%	99.96%
Source_enc	0.02%	99.98%
Avg_Price_USD	0.02%	100.00%

7. Time Series Forecasting

Global monthly deliveries were aggregated across all regions and models, producing a 132-point monthly series (Jan 2015 – Dec 2025). Two classical models were trained on data through 2023 and evaluated on the 2024 holdout.

Model	Configuration	MAE (units)	RMSE (units)	Winner
Holt-Winters	Additive trend + seasonal (period=12)	10,353	14,489	Yes
SARIMA	(2,1,2)×(1,1,1,12)	14,259	16,386	—

Holt-Winters outperforms SARIMA by 27% in MAE. Both models are re-trained on the full series for the 2026 projection.

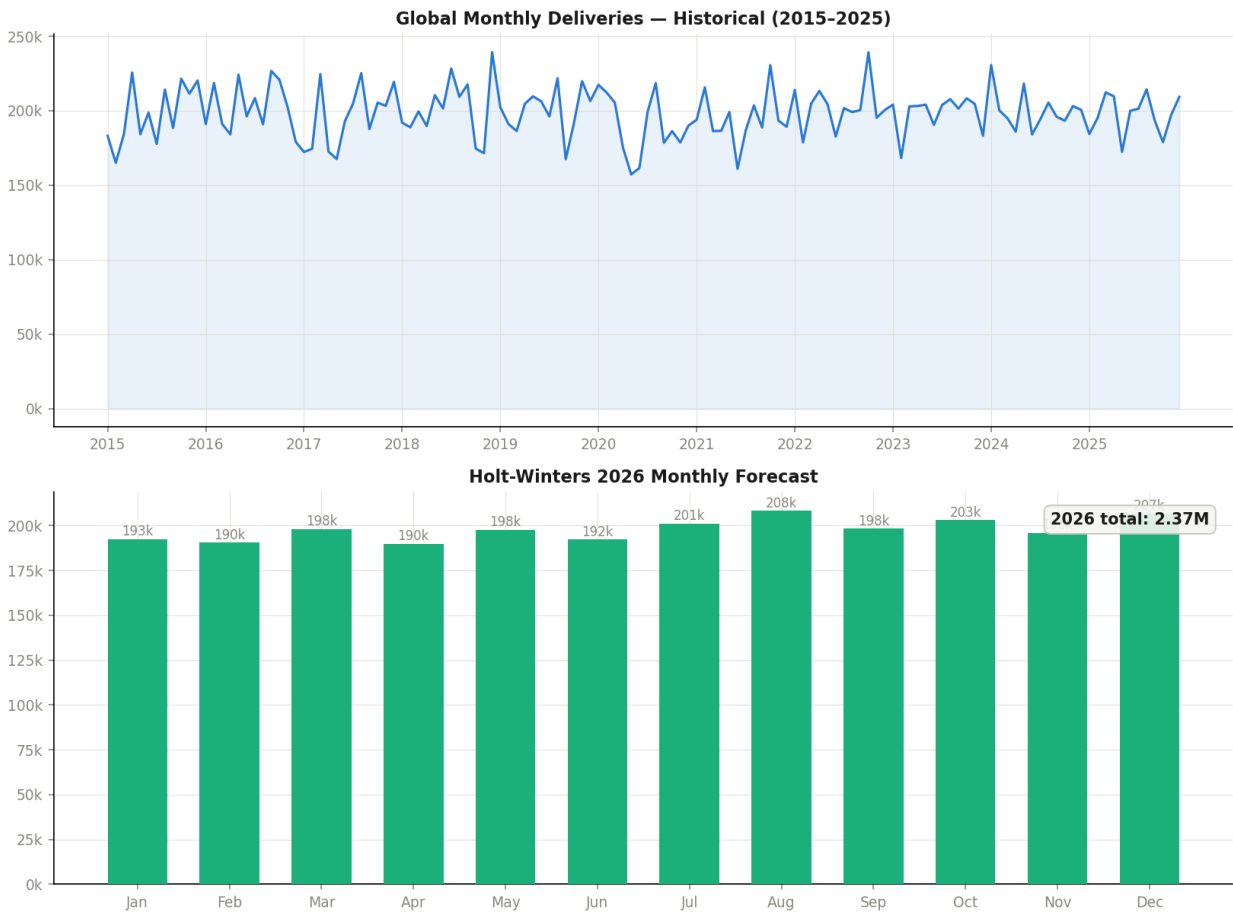


Figure 4: Historical monthly deliveries (2015–2025) and Holt-Winters 2026 monthly forecast.

2026 monthly forecast (Holt-Winters)

Jan	Feb	Mar	Apr	May	Jun
192,502	190,434	198,069	189,793	197,735	192,218
Jul	Aug	Sep	Oct	Nov	Dec
201,204	208,204	198,373	203,015	195,869	206,737

2026 projected total 2,374,153 All 12 months, global	Peak month Aug 208,204 units	Trough month Apr 189,793 units	YoY vs 2025 ~+0.2% Stable flat trend
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8. Summary & Key Takeaways

- **Production drives deliveries.** Feature importance shows `Production_Units` alone explains 96% of variance. Demand-side factors (price, range, region) are nearly irrelevant once production is known.
- **Tree models dominate linear baselines.** XGBoost and Gradient Boosting reduce MAE by ~40% vs Ridge/Lasso, confirming nonlinear structure in the fulfillment process.
- **Hyperparameter tuning gave marginal gain.** Grid search confirmed `max_depth=3`, `n_estimators=200` — the shallow depth finding is consistent with the strong single-feature dominance.
- **Holt-Winters outperforms SARIMA for this series.** The delivery series has a gentle additive seasonal pattern well-suited to exponential smoothing. SARIMA's additional AR/MA complexity adds noise.
- **2026 outlook is stable.** The Holt-Winters forecast of 2.37M global deliveries represents a flat trajectory, with mild mid-year seasonality (August peak).
- **Price is hard to predict from operational features.** The price regression R^2 of -0.27 indicates that pricing is governed by external factors (incentive programs, competitive dynamics, model refreshes) not captured in the dataset.

Report generated by Claude (Anthropic) · Data: Tesla Deliveries Dataset 2015–2025